

Lp Spaces

1.1 $\mathcal{L}^p$

Hölder's Inequality

Definition 1.1.1 ▶ $\|f\|_p$

Suppose that $(X, \mathcal{S}, \mu)$ is a measure space, $0 < p < \infty$, and $f : X \rightarrow \mathbb{F}$ is $\mathcal{S}$ -measurable. Then the ***p*-norm** of f is denoted by $\|f\|_p$ and is defined by

$$\|f\|_p = \left(\int |f|^p d\mu \right)^{1/p}.$$

Also, $\|f\|_\infty$, which is called the ***essential supremum*** of f , is defined by

$$\|f\|_\infty = \inf\{t > 0 : \mu(\{x \in X : |f(x)| > t\}) = 0\}.$$

Definition 1.1.2 ▶ $L^p(\mu)$

Suppose $(X, \mathcal{S}, \mu)$ is a measure space and $0 < p \leq \infty$. The ***Lebesgue space*** $L^p(\mu)$, sometimes denoted $L^p(X, \mathcal{S}, \mu)$, is defined to be the set of $\mathcal{S}$ -measurable functions $f : X \rightarrow \mathbb{F}$ such that $\|f\|_p < \infty$.

Theorem 1.1.3 ▶ $L^p(\mu)$ is a vector space

Suppose $(X, \mathcal{S}, \mu)$ is a measure space and $0 < p < \infty$. Then

(a)

$$\|f + g\|_p^p \leq 2^p(\|f\|_p^p + \|g\|_p^p)$$

and

(b)

$$\|\alpha f\|_p = |\alpha| \|f\|_p$$

for all $f, g \in L^p(\mu)$ and all $\alpha \in \mathbb{F}$. Furthermore, with the usual operations of addition and scalar multiplication of functions, $L^p(\mu)$ is a vector space.

Definition 1.1.4 ▶ dual exponent; p'

For $1 \leq p \leq \infty$, the **dual exponent** of p is denoted by p' and is the element of $[1, \infty]$ such that

$$\frac{1}{p} + \frac{1}{p'} = 1.$$

Lemma 1.1.5 ▶ Young's inequality

Suppose $1 < p < \infty$. Then

$$ab \leq \frac{a^p}{p} + \frac{b^{p'}}{p'}$$

for all $a \geq 0$ and $b \geq 0$.

Theorem 1.1.6 ▶ Hölder's Inequality

Suppose $(X, \mathcal{S}, \mu)$ is a measure space, $1 \leq p \leq \infty$, and $f, h : X \rightarrow \mathbb{F}$ are $\mathcal{S}$ -measurable. Then

$$\|fh\|_1 \leq \|f\|_p \|h\|_{p'}.$$

Theorem 1.1.7 ▶ $L^q(\mu) \subset L^p(\mu)$ if $p < q$ and $\mu(X) < \infty$

Suppose $(X, \mathcal{S}, \mu)$ is a finite measure space and $0 < p < q < \infty$. Then

$$\|f\|_p \leq \mu(X)^{(q-p)/(pq)} \|f\|_q$$

for all $f \in L^q(\mu)$. Furthermore, $L^q(\mu) \subset L^p(\mu)$.

Minkowski's Inequality**Theorem 1.1.8** ▶ formula for $\|f\|_p$

Suppose $(X, \mathcal{S}, \mu)$ is a measure space, $1 \leq p < \infty$, and $f \in L^p(\mu)$. Then

$$\|f\|_p = \sup \left\{ \left| \int fh \, d\mu \right| : h \in L^{p'}(\mu) \text{ and } \|h\|_{p'} \leq 1 \right\}.$$

Theorem 1.1.9 ▶ Minkowski's inequality

Suppose $(X, \mathcal{S}, \mu)$ is a measure space, $1 \leq p \leq \infty$, and $f, g \in L^p(\mu)$. Then

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

1.2 L^p **Definition of $L^p(\mu)$** **Definition 1.2.1 ▶ $\mathcal{Z}(\mu); \tilde{f}$**

Suppose $(X, \mathcal{S}, \mu)$ is a measure space and $0 < p \leq \infty$.

- $\mathcal{Z}(\mu)$ denotes the set of $\mathcal{S}$ -measurable functions from X to $\mathbb{F}$ that equal 0 almost everywhere.
- For $f \in L^p(\mu)$, let $\tilde{f}$ be the subset of $L^p(\mu)$ defined by

$$\tilde{f} = \{f + z : z \in \mathcal{Z}(\mu)\}.$$

Definition 1.2.2 ▶ $L^p(\mu)$

Suppose μ is a measure and $0 < p \leq \infty$.

- Let $L^p(\mu)$ denote the collection of subsets of $L^p(\mu)$ defined by

$$L^p(\mu) = \{\tilde{f} : f \in L^p(\mu)\}.$$

- For $\tilde{f}, \tilde{g} \in L^p(\mu)$ and $\alpha \in \mathbb{F}$, define $\tilde{f} + \tilde{g}$ and $\alpha\tilde{f}$ by

$$\tilde{f} + \tilde{g} = (f + g)^\sim \quad \text{and} \quad \alpha\tilde{f} = (\alpha f)^\sim.$$

Definition 1.2.3 ▶ $\|\cdot\|_p$ on $L^p(\mu)$

Suppose μ is a measure and $0 < p \leq \infty$. Define $\|\cdot\|_p$ on $L^p(\mu)$ by

$$\|\tilde{f}\|_p = \|f\|_p$$

for $f \in L^p(\mu)$.

Theorem 1.2.4 $\triangleright L^p(\mu)$ is a normed vector space

Suppose μ is a measure and $1 \leq p \leq \infty$. Then $L^p(\mu)$ is a vector space and $\|\cdot\|_p$ is a norm on $L^p(\mu)$.

 $L^p(\mu)$ is a Banach Space**Theorem 1.2.5**

Suppose $(X, \mathcal{S}, \mu)$ is a measure space and $1 \leq p \leq \infty$. Suppose $f_1, f_2, \dots$ is a sequence of functions in $L^p(\mu)$ such that for every $\varepsilon > 0$, there exists $n \in \mathbb{Z}^+$ such that

$$\|f_j - f_k\|_p < \varepsilon$$

for all $j \geq n$ and $k \geq n$. Then there exists $f \in L^p(\mu)$ such that

$$\lim_{k \rightarrow \infty} \|f_k - f\|_p = 0.$$

Proof sketch.

对于 $1 \leq p < \infty$:

- 取 $\sum_{k=1}^{\infty} \|f_k - f_{k-1}\|_p < \infty$ 的子列.
- 使用 Minkowski 说明绝对 $\sum_{k=1}^{\infty} |f_k - f_{k-1}|$ 的几乎处处收敛. 原级数几乎处处收敛, 等于极限 $\lim_{m \rightarrow \infty} f_m(x)$.
- 使用 Fatou's Lemma, 把 $\|f_k - f\|_p$ 中的极限函数 f 拆解, 控制成 $\liminf_{j \rightarrow \infty} \|f_k - f_j\|_p$ 以带入 Cauchy 条件.